tf.compat.v1.nn.static_rnn Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/static_rnn

> *Creates a recurrent neural network specified by RNNCell cell. (deprecated)*

Function Signatures

```
tf.compat.v1.nn.static_rnn( cell, inputs, initial_state=None, dtype=None, sequence_length=None, scope=None )
```

```
keras.layers.RNN(cell, unroll=True)
```

```
state = cell.zero_state(...) outputs = [] for input_ in inputs: output, state = cell(input_, state) outputs.append(output) return (outputs, state)
```

Code Examples

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tf.compat.v1.nn.static_rnn(
cell,
inputs,
initial_state=None,
dtype=None,
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```
(output, state)(b, t) =
(t >= sequence_length(b))
? (zeros(cell.output_size), states(b, sequence_length(b) - 1))
: cell(input(b, t), state(b, t - 1))
```


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(output, state)(b, t) =
(t >= sequence_length(b))
? (zeros(cell.output_size), states(b, sequence_length(b) - 1))
: cell(input(b, t), state(b, t - 1))
```

Documentation

Creates a recurrent neural network specified by RNNCell cell. (deprecated)

The simplest form of RNN network generated is:

However, a few other options are available:

An initial state can be provided.
If the sequence_length vector is provided, dynamic calculation is performed.
This method of calculation does not compute the RNN steps past the maximum
sequence length of the minibatch (thus saving computational time),
and properly propagates the state at an example's sequence length
to the final state output.

The dynamic calculation performed is, at time t for batch row b,

Returns

• outputs is a length T list of outputs (one for each input), or a nested
tuple of such elements.

- state is the final state

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Raises